



# OPEN A cycle-aware and physics-informed framework for battery remaining useful life prediction

Yixuan Chen<sup>1</sup>, Yueran Wu<sup>2</sup> & Conghui Li<sup>1</sup>✉

Accurate prediction of the Remaining Useful Life (RUL) of Li-ion batteries is essential for safe and efficient energy systems, but it is difficult because sensor data are multivariate and irregularly sampled. Many state-of-the-art deep learning methods are domain-agnostic. For example, T-PATCHGNN uses generic patching that cuts charge–discharge cycles into fragments with little physical meaning, weakening the input signal and limiting learning of true degradation. We propose Bat-T-GNN, which injects domain knowledge at both the input and objective levels. First, Cycle-Aware Patching segments the time series by actual charge–discharge cycles, giving the model coherent, physically meaningful inputs. Second, a Physics-Informed Consistency Loss (PINN-RUL) regularizes training by requiring the final RUL prediction to be consistent with a physically plausible degradation curve learned from the data. This method experimented on public benchmarks show that proposed method significantly outperforms prior methods, including T-PATCHGNN. Ablation studies confirm that both proposed components drive the performance gains, establishing a new state of the art in battery RUL prediction.

Lithium-ion batteries power many modern systems, from electric vehicles to grid storage. Safe and cost-effective operation requires accurate prediction of Remaining Useful Life (RUL)<sup>1,2</sup>. This is difficult because RUL is a time-series forecasting problem in which battery health declines due to many interacting factors<sup>3–6</sup>.

The main challenge comes from the properties of real battery data. First, sampling within operating cycles is inconsistent and irregular. In practice, a Battery Management System (BMS) does not measure at a fixed rate: sampling can be denser during rapid changes and sparser during stable or resting periods<sup>7–9</sup>. This violates the uniform-grid assumption used by many standard models such as CNNs and Transformers. Moreover, the time gap between two measurements is informative, such as response rates, so simple fixes like linear interpolation can add artificial points and hide true non-linear behavior. Second, RUL prediction must model complex, evolving relationships among multiple variables, including voltage, current, temperature, and internal resistance. No single sensor captures battery health; it emerges from their joint behavior. These correlations change as the battery ages. For example, internal resistance may have little effect early on, but it increases with degradation and then strongly drives voltage drop and heat generation near end-of-life. Thus, a useful model cannot assume fixed correlations. It must adapt to stage-dependent interactions to predict critical transitions, including the late “knee-point” decline.

Historically, battery prognostics relied on physics-based models that simulate the underlying electrochemical processes<sup>10–14</sup>. While offering deep theoretical insight, many state-of-the-art deep learning methods are not well suited to this problem. Transformer models work well for general time-series forecasting, but they assume regularly sampled data and struggle with the irregular time gaps common in battery datasets<sup>15–17</sup>. Pre-processing to force regular sampling can introduce errors and hide important details. Graph Neural Networks (GNNs) can model multiple variables together, but they often assume fixed relationships among sensors<sup>18–20</sup>. This is unrealistic for aging batteries, where the importance of factors such as internal resistance increases over time. Methods based on Neural Ordinary Differential Equations (ODEs) can handle irregular sampling, but they are often too slow for practical use<sup>21,22</sup>. The Transformable Patching Graph Neural Network (T-PATCHGNN) is a recent model that addresses both irregular data and changing correlations<sup>23</sup>. However, it has a key limitation for battery applications: it splits time series into generic, equal-duration windows that ignore the natural charge–discharge cycle. This cycle is the basic unit of battery operation. By fragmenting it, the model loses meaningful structure and struggles to learn the true degradation process<sup>24,25</sup>.

To bridge this critical gap, we introduce Cycle-Aware Patching. This strategy replaces the domain-agnostic segmentation used by models like T-PATCHGNN with a physically-informed approach. Instead of dividing

<sup>1</sup>School of Information Technology, Monash University, 47500 Subang Jaya, Malaysia. <sup>2</sup>College of Information Science and Technology, Hangzhou Normal University, Hangzhou 311121, China. ✉email: Conghui.Li@monash.edu

the time series into windows of standard temporal duration, our method intelligently identifies the natural boundaries of the charge-discharge cycles by monitoring the battery's current profile. This ensures that each data patch fed into the model is a semantically complete and physically meaningful feature, representing one full operational event. This simplifies the learning task for the powerful T-PATCHGNN backbone, as it no longer needs to infer physical context from fragmented data, but can instead directly model the degradation relationship between sequential cycles. However, even with a physically coherent input, one more challenge remains. Models trained with a standard objective like Mean Squared Error (MSE) are only constrained at their final output, treating the internal feature extraction process as an unconstrained black box. This can lead to solutions that are numerically accurate for the training distribution but are not physically plausible, thus failing to generalize under different conditions. To address this lack of physical constraint, we propose our second major contribution: the Physics-Informed Consistency Loss (PINN-RUL). This multi-component loss function acts as a regularizer by imposing a physical inductive bias on the learning process. It compels the model not only to predict the final RUL but also to learn the parameters of a physically-grounded degradation curve that fits the historical data. Crucially, it then penalizes any inconsistency between the RUL prediction and the RUL implied by this learned physical trajectory. This consistency check keeps the model's internal reasoning logical and reliable. It prevents the model from relying on random or misleading correlations and instead guides it toward solutions that are both accurate and physically meaningful, improving its generalization and robustness.

Our main contributions can be summarized as follows:

1. Cycle-aware patching: we segment data by actual charge-discharge cycles, aligning inputs with battery physics to provide more meaningful signals for deep models.
2. Physics-informed consistency loss (PINN-RUL): a regularizer that enforces physically plausible predictions and improves robustness.
3. Bat-T-GNN framework: an end-to-end system that integrates these components and sets a new state of the art for RUL prediction in extensive experiments.

## Related work

The prediction of RUL for Lithium-ion batteries has seen a significant shift towards data-driven methodologies, with deep learning emerging as the dominant approach due to its ability to model complex, non-linear degradation dynamics.

The initial wave of deep learning applications for RUL prediction was led by Recurrent Neural Networks (RNNs)<sup>7,8</sup>, particularly LSTM<sup>7,9</sup> and GRU networks<sup>26</sup>. These models are inherently suited for processing sequential data, making them a natural fit for modeling the evolution of battery health indicators over consecutive charge-discharge cycles. To enhance the performance of these models, recent studies have incorporated attention mechanisms. An attention-based LSTM, for example, can adaptively assign different weights to different cycles in the battery's history, allowing the model to focus on more informative periods of degradation and leading to more accurate RUL estimations<sup>27,28</sup>. Concurrently, CNNs have been successfully employed to automate the feature extraction process. By applying 1D convolutions directly to raw sensor curves (e.g., voltage, current, temperature) from individual cycles, CNNs can learn to identify salient morphological features like voltage plateaus, peaks, and valley shapes, which are strong indicators of battery health, thereby bypassing the need for manual feature engineering<sup>29,30</sup>. A highly prevalent trend in recent years has been the development of hybrid architectures that combine these two approaches. A typical CNN-LSTM model first uses a CNN module to extract a rich, high-level feature vector from each cycle's raw data, and an LSTM module then processes the sequence of these feature vectors to model the long-term temporal dependencies of battery degradation, effectively capturing both local and global dynamics<sup>29–31</sup>.

While RNN-based models are powerful, they can struggle with modeling very long-term dependencies due to issues like vanishing gradients. To address this, Transformer-based models have recently been introduced to the field of battery prognostics<sup>15–17,32,33</sup>. The self-attention mechanism at the core of the Transformer allows the model to directly capture the influence of early-life cycles on late-life degradation, effectively relating cycles that are hundreds of operational periods apart. This provides a more global perspective on the battery's entire operational history, which is crucial for accurate long-term RUL prediction. However, these models typically require regularly sampled cycle-level data, forcing practitioners to perform feature engineering or resampling that can obscure fine-grained intra-cycle dynamics.

Another significant recent advancement involves the use of GNNs to model inter-dependencies. The majority of research in this area has focused on modeling inter-cell relationships within a battery pack. By representing each cell as a node in a graph and the electrical or thermal connections as edges, GNNs can effectively propagate health information between cells. This allows the model to account for cell-to-cell variations and interactions, such as thermal runaway or imbalanced aging, which are critical for accurate pack-level RUL prediction<sup>34–37</sup>. Furthermore, cutting-edge approaches like Physics-Informed Neural Networks have been explored to bridge the gap between purely data-driven and physics-based models. These models integrate domain knowledge by embedding simplified electrochemical equations into the network's loss function or architecture, guiding the model towards physically consistent and more generalizable RUL predictions<sup>38–41</sup>. While GNNs have proven effective for modeling inter-cell dependencies in a pack, their application to modeling the synergistic relationships between different sensor streams within a single cell remains largely unexplored.

Despite these advancements, two critical research gaps remain. Firstly, the vast majority of existing deep learning models implicitly assume that the data is regularly sampled and do not explicitly address the challenge of irregular time intervals that are common in real-world battery operational logs. Methods that ignore the actual time gaps between observations or rely on simple resampling can fail to accurately model the true rate of degradation. Secondly, while GNNs have been used to model inter-cell relationships in a pack, the rich

web of correlations between different sensor parameters within a single battery (e.g., how internal resistance correlates with temperature and  $dV/dt$  over time) remains largely underexplored. Existing models often treat these multivariate inputs as a flat feature vector, rather than as nodes in a dynamic, interacting system, thereby missing an opportunity to model the synergistic effects of degradation. Our work is positioned to address these gaps by proposing a framework that inherently handles irregular time-series data while dynamically modeling the inter-parameter relationships for more accurate and robust RUL prediction.

## Methodology

This section details the architecture and components of our proposed Bat-T-GNN model. We first provide a high-level overview of the framework and then elaborate on each key module.

### Overall framework

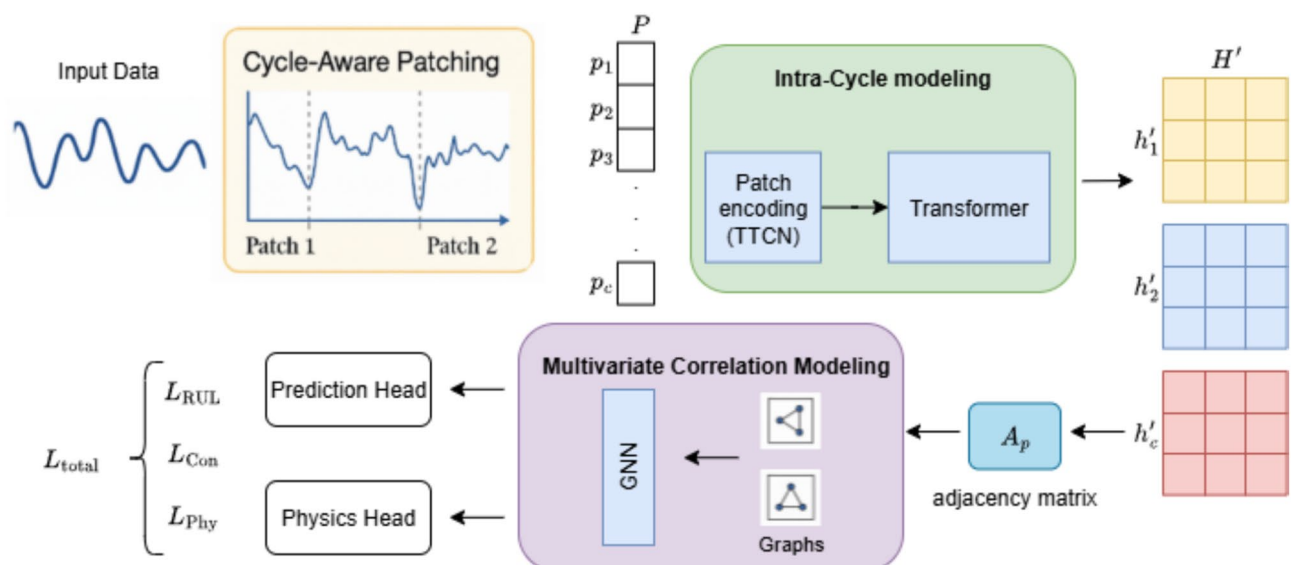
The proposed Bat-T-GNN is a comprehensive framework designed to accurately predict battery RUL from irregular multivariate time series. An overview of the model architecture is illustrated in Fig. 1. The framework processes the data in three main stages: patching, encoding, and prediction.

First, the framework receives the raw, irregularly-sampled multivariate time series as input. This data stream is immediately processed by our novel Cycle-Aware Patching module. This initial and critical step transforms the continuous data into a structured sequence of patches, where each patch corresponds to a physically meaningful charge-discharge cycle. Second, this sequence of cycle patches is fed into a powerful encoder, which is adapted from the state-of-the-art T-PATCHGNN architecture. The encoder operates in three phases to build a rich feature representation: First, a Transformable Time-aware Convolutional Network (TTCN) captures local features from the irregular data points within each cycle patch. Next, a Transformer module models the long-term dependencies between the cycles, learning the battery's overall degradation trend over its lifetime. Finally, a time-varying adaptive Graph Neural Network (GNN) explicitly models the dynamic correlations between the different sensor variables (e.g., voltage, current, temperature). Finally, the comprehensive representation generated by the encoder is passed to a simple yet effective output layer, which maps the learned features to the final RUL prediction. The following sections will provide a detailed description of each component.

### Cycle-aware patching

The central premise of our work is that even the most powerful deep learning models are fundamentally “domain-agnostic.” They learn statistical patterns from the data they are given, but they have no innate understanding of the physical processes that generate that data. When a generic model like T-PATCHGNN encounters a long stream of battery data, its default patching mechanism—which segments data into chunks of uniform time duration or length—acts like someone tearing a book into pages of exactly 100 words. Inevitably, sentences will be split in half, destroying their meaning. Similarly, A generic patching method can arbitrarily split a single discharge event across two separate patches, corrupting the semantic integrity of the data and making it difficult for the model to learn the true relationship between operational work and health degradation. This forces the downstream model to solve two incredibly difficult problems at once: first, it must try to mentally reassemble the broken pieces of physical events, and second, it must predict future degradation based on this scrambled information.

Our Cycle-Aware Patching strategy is designed to solve this fundamental problem by acting as an intelligent interface between the raw physical world and the abstract world of the deep learning model. We argue that for batteries, the most indivisible and meaningful unit of operation is the charge-discharge cycle. Each cycle

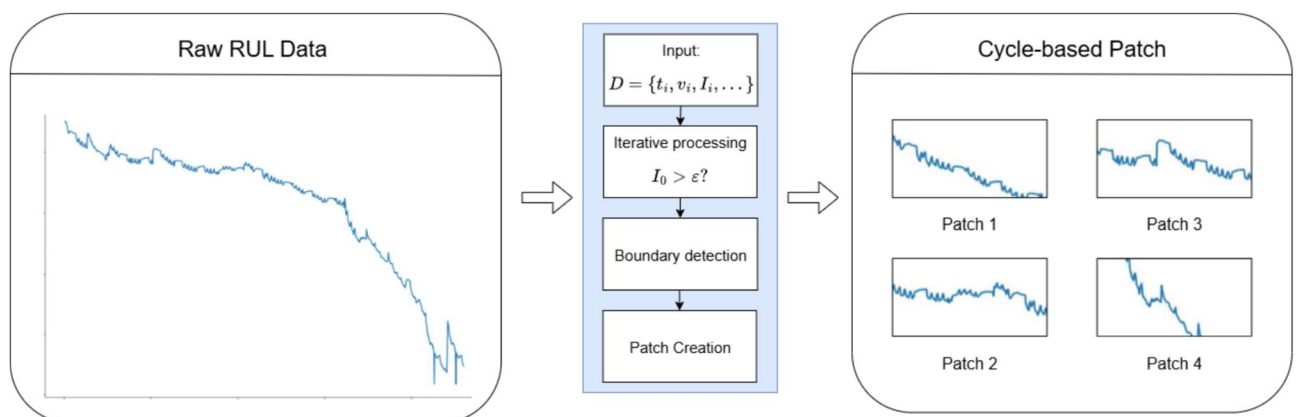


**Fig. 1.** Structure of methodology.

represents a complete unit of work that contributes a small amount of irreversible degradation to the battery's health. By ensuring that each patch fed to our model corresponds to exactly one of these cycles, we provide the network with clean, coherent, and physically consistent inputs.

To implement this strategy, we developed a robust algorithm that identifies cycle boundaries by monitoring the battery's current flow, a signal universally available in battery datasets. The logic, detailed in Algorithm 1 and Fig. 2, can be broken down into the following steps:

1. **Initialization:** The algorithm begins by creating an empty list to hold the final Patches and a temporary list namely current patch, to accumulate data for the cycle currently being processed. It also determines the initial phase (either CHARGING or DISCHARGING) based on the very first current measurement,  $I_0$ , establishing the starting context.
2. **Iterative accumulation:** The algorithm then iterates through every data point  $d_i$  in the raw time series. For each point, it performs two simple actions: it appends the data point to the current patch, steadily building up the record of the ongoing cycle, and it determines the current phase by checking if the current  $I_i$  is positive (charging) or negative (discharging).
3. **Boundary detection:** The most critical step occurs inside the main loop. The algorithm constantly checks for a specific state transition: a change from a DISCHARGING phase to a CHARGING phase. This specific transition is a highly reliable indicator that a full operational cycle (which includes a full discharge) has just been completed. This event serves as the delimiter for our patches.
4. **Patch creation and reset:** When this transition is detected, the algorithm knows a complete cycle has been captured. It appends the entire current patch (which now contains all the data points from that one cycle) to the main Patches list. Immediately after, it resets current patch to an empty list, ready to start accumulating data for the next cycle. This ensures a clean separation between consecutive cycles.
5. **Handling the final patch:** An important edge case is handled after the loop finishes. The final cycle in the dataset might not end with a perfect transition (the data logging may simply stop). The final check ensures that any data accumulated in current patch during the last, potentially incomplete, cycle is also saved as the final patch.



**Fig. 2.** Overview of cycle-aware patching.

**Input:** Raw multivariate time series  $D = \{(t_i, v_i, I_i, \dots)\}$ , where  $I_i$  is the current.

A near-zero current threshold  $\varepsilon$ .

**Output:** A list of cycle-based patches  $P = \{p_1, p_2, \dots, p_C\}$ .

```
// Initialization
Patches  $\leftarrow \emptyset$ 
current_patch  $\leftarrow \emptyset$ 
if  $I_0 > \varepsilon$  then
  | phase  $\leftarrow$  CHARGING
end
else
  | phase  $\leftarrow$  DISCHARGING
end

for each data point  $d_i = (t_i, v_i, I_i, \dots)$  in  $D$  do
  current_patch.append( $d_i$ )
  // Determine the current phase
  if  $I_i > \varepsilon$  then
    | current_phase  $\leftarrow$  CHARGING
  end
  else
    | current_phase  $\leftarrow$  DISCHARGING
  end
  // A transition from DISCHARGING to CHARGING marks the end of a full cycle
  if phase == DISCHARGING and current_phase == CHARGING then
    | Patches.append(current_patch)
    | current_patch  $\leftarrow \emptyset$  // Start a new patch for the next cycle
  end
  phase  $\leftarrow$  current_phase
end
// Add the last remaining patch if it exists
if current_patch is not empty then
  | Patches.append(current_patch)
end
return Patches
```

#### Algorithm 1. Cycle-aware patching.

### Encoder architecture

The encoder serves as the core analytical engine of our framework. Its primary role is to process the sequence of cycle-aware patches and learn a rich, comprehensive feature representation that captures the battery's complex degradation dynamics. For this purpose, we employ the sophisticated and mature architecture of the T-PATCHGNN framework [cite], which has demonstrated outstanding performance on general irregular time-series tasks. Our contribution is not to redesign this powerful encoder, but rather to provide it with the physically meaningful inputs it requires to operate at its full potential. The encoder is composed of three key modules that work in sequence to model dependencies at different levels of abstraction.

#### Intra-cycle feature extraction

The initial output of our Cycle-Aware Patching strategy is a sequence of patches,  $P = p_1, p_2, \dots, p_C$ , where each patch represents a complete physical cycle. The first challenge within the encoder is to process each of these patches and summarize the complex dynamics occurring within that single operational cycle into a single, fixed-size feature vector  $h_c$ . This task is non-trivial, as each patch contains a variable number of multivariate observations recorded at irregular time intervals. Standard deep learning tools are ill-suited for this structure. Conventional Convolutional Neural Networks (CNNs) require data to be on a regular grid, while Recurrent Neural Networks (RNNs), though capable of handling variable lengths, process data sequentially and may struggle to capture the holistic features of an entire cycle event at once.

To address this challenge, we employ the Transformable Time-aware Convolution Network (TTCN), a sophisticated module introduced in the T-PATCHGNN framework. TTCN is engineered to perform a convolution-like operation directly on irregular time series. Its core innovation lies in its use of a meta-learning paradigm. Instead of applying a single, fixed-size filter to all patches, TTCN learns a meta-filter that dynamically generates a unique, custom-sized convolution filter for every individual patch it processes. This adaptability

makes it exceptionally well-suited for our sequence of physically-derived yet temporally-irregular cycle-patches. The process of encoding a patch with TTCN begins by representing the raw, continuous timestamps as vectors using a continuous time embedding function, which captures both periodic and non-periodic temporal patterns. This embedded patch data is then passed to the meta-filter, which generates the parameters for a larger, one-dimensional convolution filter whose size is tailored to perfectly match the length of that specific patch. This custom-generated, “time-aware” filter is then convolved across the patch’s data points. This single operation allows the module to holistically capture the local semantics and patterns across the entire cycle. To ensure that the output magnitudes are consistent across patches of different lengths and data densities, the derived filter parameters are normalized along the temporal dimension. Through this procedure, TTCN transforms each irregular cycle-patch  $p_c$  into a robust, fixed-dimension embedding vector  $h_c$ . The final result of this stage is the conversion of the entire sequence of irregular patches  $P$  into a regular, structured sequence of feature embeddings,  $H = (h_1, h_2, \dots, h_C)$ . This clean sequence is now perfectly prepared for the subsequent module, which will model the long-range dependencies between these cycle embeddings.

### Inter-cycle dependency modeling

Following the intra-cycle feature extraction, our model has a structured sequence of embeddings,  $H = (h_1, h_2, \dots, h_C)$ , where each vector represents a single operational cycle. The next critical task is to model the long-range temporal dependencies that exist between these cycles. The degradation of a battery is a path-dependent and cumulative process; its state of health at cycle  $C$  is not merely a function of cycle  $C-1$ , but is influenced by the entire operational history. A sudden high-load event at an early cycle, for instance, can initiate a degradation pathway that only becomes fully apparent hundreds of cycles later. Capturing these long-term, non-linear relationships is therefore essential for accurate RUL prediction. While Recurrent Neural Networks (RNNs) and their variants like LSTMs have traditionally been used for sequence modeling, they often struggle with very long sequences, such as the thousands of cycles in a battery’s lifetime, due to the well-known vanishing gradient problem. To effectively capture these long-range dependencies, we employ a standard Transformer Encoder<sup>42</sup>, which has become the state-of-the-art for sequence modeling tasks. The sequence of cycle embeddings  $H$  is treated as a sequence of input tokens for the Transformer. At the heart of the Transformer is the self-attention mechanism, which enables the model to weigh the relative importance of all cycles in the sequence when updating the representation for any single cycle. This mechanism creates direct pathways between any two points in the sequence, allowing the model to learn, for example, that the characteristics of cycle 50 are highly relevant for understanding the state at cycle 500, overcoming the sequential bottleneck of RNNs. The output of the Transformer module is a new sequence of context-aware embeddings,  $H' = (h'_1, h'_2, \dots, h'_C)$ . Each vector  $h'_c$  in this sequence is a more sophisticated representation than its input counterpart. It not only summarizes the events of cycle  $c$  but is also imbued with the historical context of the entire degradation trajectory that led up to it. This temporally-enriched representation is crucial, as it provides the necessary context for the final module of the encoder, which models the state of the multivariate correlations at that specific point in the battery’s life.

### Dynamic multivariate correlation modeling

The final task of the encoder is to explicitly model the relationships between the different sensor variables. The Transformer module successfully captured the degradation history along the time axis, but a battery’s state is also a function of the instantaneous, synergistic interactions between its various physical parameters. A critical aspect of battery aging is that these correlations are not static. For example, the influence of internal resistance on capacity fade is relatively minor in the early stages of life but becomes a dominant and highly correlated indicator of failure as the battery approaches its end of life. A model that assumes fixed relationships between variables would miss these crucial dynamic shifts. To capture this dynamic behavior, we utilize the time-varying adaptive Graph Neural Network (GNN) component from the T-PATCHGNN framework. This approach moves beyond traditional methods that use a single, predefined graph structure (e.g., based on sensor proximity) or learn one static graph for the entire dataset. Instead, this module dynamically constructs a unique graph, represented by a cycle-specific adjacency matrix  $A_c$ , for each individual cycle-patch  $p_c$ . This allows the model to represent the precise state of multivariate correlations at that specific point in the battery’s life. The generation of this dynamic graph is a learned process guided by an attention mechanism. This mechanism synthesizes information from two distinct sources: first, a set of learnable, static embeddings that capture the inherent, time-invariant characteristics of each variable (for instance, the fundamental electrochemical link between voltage and current). Second, it uses the context-aware cycle embeddings  $h'_c$  from the Transformer module, which provide the dynamic, time-varying signal reflecting the battery’s state at that particular cycle. By combining these static and dynamic representations, the model learns a unique set of pairwise relationships that constitute the adjacency matrix  $A_c$  for that cycle. Once this bespoke graph structure  $A_c$  is established for the cycle, a standard Graph Neural Network layer is applied. The GNN uses the learned graph to propagate and aggregate information across the different variable nodes. This step effectively fuses the features of correlated variables, allowing the model to learn complex, multivariate patterns. For example, it can learn that at a late stage of life, a subtle change in the voltage curve is significantly more alarming when it occurs simultaneously with a sharp rise in temperature, because the learned graph  $A_c$  for that cycle has established a strong edge between those two nodes. The final output of this module is a sequence of highly refined embeddings that have integrated intra-cycle dynamics, inter-cycle historical context, and dynamic inter-variable correlations, thus completing the encoder’s comprehensive analysis of the battery’s state.

### Physics-informed consistency loss

The final stage of the Bat-T-GNN framework is to translate the feature representations learned by the encoder into a final, accurate RUL prediction. This is achieved through a specialized prediction architecture and a novel loss function designed to ensure that the model's predictions are not only data-driven but also physically plausible.

While the mean squared error (MSE) is a standard loss function for regression, it has a key limitation in this context: it is “physics-agnostic.” An MSE-based loss treats the encoder as a black box, only penalizing the error on the final RUL. It has no mechanism to verify if the degradation *trajectory* implied by the model's learned features is physically sensible. To address this, we introduce a new multi-component loss function, the *Physics-Informed Consistency Loss for RUL (PINN-RUL)*, which injects domain knowledge directly into the training objective. To facilitate this, we augment our model with two parallel output heads: a main *Prediction Head* that directly predicts the final RUL, and a *Physics Head* that predicts the parameters of a well-established empirical model for battery capacity fade, such as a double-exponential decay curve.

Our total loss function,  $L_{\text{total}}$ , is a weighted sum of three distinct components: a primary RUL data fidelity loss, a physics-informed shape loss, and a crucial prediction consistency loss, as shown in Eq. (1).

$$L_{\text{total}} = L_{\text{RUL}} + \lambda_{\text{phys}} \cdot L_{\text{phys}} + \lambda_{\text{cons}} \cdot L_{\text{cons}} \quad (1)$$

where  $\lambda_{\text{phys}}$  and  $\lambda_{\text{cons}}$  are hyperparameters that balance the contribution of each term.

The first component, the *RUL Data Fidelity Loss* ( $L_{\text{RUL}}$ ), is the standard Mean Squared Error on the final RUL prediction. Its purpose is to ensure the model remains sharply focused on its primary objective of predicting the correct end-of-life point, as defined in Eq. (2).

$$L_{\text{RUL}} = (\text{RUL}_{\text{pred}} - \text{RUL}_{\text{true}})^2 \quad (2)$$

where  $\text{RUL}_{\text{pred}}$  is the direct output from the main Prediction Head.

The second component, the *Physics-Informed Shape Loss* ( $L_{\text{phys}}$ ), forces the model's learned representation to also understand the *shape* of the degradation process. It uses the parameters output by the Physics Head to construct an empirical capacity curve,  $C_{\text{phys}}(t)$ , and then penalizes the deviation of this curve from the true historical capacity data observed across  $C$  cycles (Eq. 3).

$$L_{\text{phys}} = \frac{1}{C} \sum_{i=1}^C (C_{\text{phys}}(t_i) - C_{\text{true}}(t_i))^2 \quad (3)$$

The most innovative component is the *Prediction Consistency Loss* ( $L_{\text{cons}}$ ), which enforces logical consistency between the model's two parallel outputs. The main head predicts the RUL directly ( $\text{RUL}_{\text{pred}}$ ), while the Physics Head implies an RUL ( $\text{RUL}_{\text{phys}}$ ) based on its predicted decay curve. This loss penalizes any divergence between these two predictions (Eq. 4), acting as a powerful regularizer that discourages “unphysical” RUL estimates that are inconsistent with the observed degradation trend.

$$L_{\text{cons}} = (\text{RUL}_{\text{pred}} - \text{RUL}_{\text{phys}})^2 \quad (4)$$

The simultaneous optimization of these three components transforms the learning objective. Instead of merely fitting a single RUL value, the model is now constrained to also learn the parameters of the underlying physical degradation process. The consistency loss  $L_{\text{cons}}$ , ensures that the final RUL prediction is a logical consequence of this learned trajectory. This methodology imposes a strong physical inductive bias on the model, guiding it towards solutions that are not only numerically accurate but also physically meaningful, thereby enhancing its generalization capabilities and robustness.

## Experiment

### Dataset

In the field of battery RUL prediction, the availability of comprehensive and well-documented datasets is crucial for developing and validating accurate prognostic models. In this study, the NASA Prognostics Center of Excellence (PCoE) and the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland provide foundational datasets. Although some updated datasets have emerged in recent years. However, due to the extensive use of NASA and CALCE datasets, they still have a high degree of relevance in verifying the model's ability to capture fundamental degradation mechanisms under clear and repeatable conditions and in comparing performance with other sota models under fair conditions.

### NASA dataset

The NASA PCoE offers several key datasets focused on 18650-format Lithium-ion batteries<sup>43</sup>. A frequently used dataset involves 34 cells that were cycled until they reached 70% or 80% of their initial 2Ah capacity. These experiments included charging, discharging, and electrochemical impedance spectroscopy at various temperatures. Another significant collection is the Randomized Battery Usage Dataset, which contains data from 28 Li-ion cells, divided into seven groups. Five of these groups were tested at room temperature, and two were tested at 40°C. A well-known subset of the NASA data, often used in prognostic studies, includes batteries B0005, B0006, B0007, and B0018. These specific 18650 Li-ion cells had a rated capacity of 2Ah and were subjected to consistent charging at 1.5A to 4.2V. However, they underwent different discharging protocols to a

constant current of 2A, with voltage cutoffs of 2.7V, 2.5V, 2.2V, and 2.5V respectively. More recently, a dataset focusing on accelerated life testing was released, which includes data from 26 battery packs, each composed of two 18650 cells, tested under both constant and random loading conditions. The sensor data for the NASA dataset was recorded at irregular intervals during the operational cycles. Sampling frequencies can vary, with denser measurements during periods of rapid change and sparser data during rest periods. This temporal inconsistency necessitates a method like our Continuous Time Embedding to properly contextualize each data point in time, preventing the loss of information that would occur with simple resampling. In this experiment, We selected Output Current (Amps), Loaded Voltage(V) and Temperature of Battery( $^{\circ}$ C) as input.

### CALCE dataset

The CALCE at the University of Maryland provides a more diverse collection of battery data, covering a wide range of chemistries, designs, and test conditions<sup>44</sup>. Their repository includes data from cylindrical, pouch, and prismatic cells with chemistries such as Lithium Cobalt Oxide (LCO), Lithium Iron Phosphate (LFP), and Nickel Manganese Cobalt Oxide (NMC). Specific experiments provide detailed numerical data. For example, the CS2 dataset contains results from 15 prismatic LCO cells cycled at room temperature. In another extensive test, a total of 144 Li-ion cells were used for a storage life study, examining the effects of three different state of charge (SOC) levels and four different temperatures. CALCE also offers data from experiments designed to quantify the impact of partial charge-discharge cycling on pouch cells. The availability of varied data, including dynamic driving profiles, makes the CALCE dataset a comprehensive resource for evaluating the general applicability of RUL prediction models. In total, some studies have utilized up to 179 cells from various sources, including CALCE, to develop and test machine learning pipelines for State of Health (SOH) estimation. Similar to the NASA dataset, the CALCE data also suffers from significant temporal inconsistencies. The data logging does not strictly follow a fixed sampling frequency. In this experiment, We selected  $dV/dt(V/s)$ , Internal Resistance(Ohm), Discharge Energy(Wh) and Charge Capacity(Ah) as input. The voltage change rate ( $dV/dt$ ) was chosen as it can be a more sensitive indicator of certain degradation modes than the absolute voltage level. For instance, changes in the slope of the voltage curve during discharge can reveal alterations in the battery's internal kinetics.

### UNIBO dataset

The UNIBO Powertools dataset<sup>45</sup> comprises experimental data from 27 Li-ion batteries, encompassing a diverse range of cell types and operating conditions. The standard experimental protocol involved charging the cells at a constant current of 1.8 A to a cutoff voltage of 4.2 V, followed by a constant current discharge at 5 A until a predefined end-of-discharge voltage ( $V_{eod}$ ). Reference cycles to measure key health indicators like capacity and resistance were performed every 100 cycles. A key feature of this dataset is its heterogeneity; the batteries are categorized into seven distinct groups based on several factors: the manufacturer, the intended cell application, and the specific test profile. These profiles include a standard 5 A discharge, a high-current 8 A discharge, and a pre-conditioned test involving 90 days of storage at  $45^{\circ}$ C prior to cycling. This diversity in cell types and stress conditions makes the dataset particularly well-suited for evaluating the generalization capabilities of RUL prediction models. In this experiment, we selected temperature, voltage, current, Wh measured during charge ( $wh\_charging$ ) and Wh measured during discharge ( $wh\_discharging$ ).

### Evaluation metrics

To comprehensively assess the performance of our proposed RUL prediction model and compare it against baseline methods, we employ standard regression evaluation metrics. These metrics provide insights into the accuracy and reliability of the predicted RUL values. For each battery in the test set, we calculate the following metrics:

**Root mean squared error (RMSE):** RMSE quantifies the difference between predicted and actual RUL values by taking the square root of the mean of squared errors. It penalizes larger errors more heavily, providing a measure of the magnitude of errors.

$$RMSE = \sqrt{\frac{1}{N_{test}} \sum_{i=1}^{N_{test}} (y_i - \hat{y}_i)^2}$$

where  $y_i$  is the true RUL and  $\hat{y}_i$  is the predicted RUL for the  $i$ -th test instance.

**Mean absolute error (MAE):** MAE measures the average magnitude of the errors between predicted and actual RUL values. It provides a more direct interpretation of the average error size, as it is not squared.

$$MAE = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} |y_i - \hat{y}_i|$$

**Mean absolute percentage error (MAPE):** MAPE expresses the forecast error as a percentage of the actual RUL, offering a relative measure of accuracy that is less sensitive to the scale of the RUL values.

$$MAPE = \frac{100\%}{N_{test}} \sum_{i=1}^{N_{test}} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Note: MAPE can be sensitive to true RUL values close to zero. For RUL prediction, it's typically computed only for instances where  $y_i > 0$ .

Performance analysis

This section presents a comprehensive evaluation of our proposed model against several state-of-the-art baselines on the NASA, CALCE and UNIBO battery datasets. We compare our method with MC-CNN-TimesNet<sup>46</sup>, DTNN<sup>32</sup>, Conditional GCN<sup>47</sup> and T-PATCHGNN<sup>23</sup>. All models were trained and evaluated under consistent conditions to ensure a fair comparison.

The performance of all models is summarized in Table 1, reporting RMSE, MAE, and MAPE. We highlight the best-performing models with bold text and the second-best with underline.

As illustrated in Table 1, our proposed model, Bat-T-GNN, consistently outperforms all baseline methods across both the NASA, CALCE and UNIBO datasets. Specifically, on the NASA dataset, our model achieves the lowest RMSE of 0.014, the MAE of 0.007, and the lowest MAPE of 0.008. Similarly, on the CALCE dataset, Bat-T-GNN demonstrates superior performance with an RMSE of 0.024, an MAE of 0.011, and a MAPE of 0.012. These results demonstrate the superior accuracy of our model in predicting the Remaining Useful Life (RUL). The significant performance gains are primarily attributed to the effective handling of irregular sampling through transformable patching and the accurate capture of dynamic inter-parameter correlations via attention-based GNNs.

The baseline models, particularly those that do not explicitly handle irregular time intervals or fail to leverage multivariate information effectively, exhibit significantly higher errors. For instance, DTNN still shows higher errors across all metrics on both datasets. Its RMSE of 0.026 on the NASA dataset and 0.039 on the CALCE dataset highlights the performance gap. Furthermore, methods like conditional GCN, which yield an RMSE of 0.041 on the NASA dataset and 0.079 on the CALCE dataset, struggle to capture the complex, multi-faceted degradation dynamics evident in battery data. Our approach's ability to model dynamic inter-parameter relationships, as enabled by the attention-based graph structure learning, proves crucial in distinguishing nuanced degradation patterns.

Case study

To provide a qualitative assessment of our model's performance, Figs. 3 and 4 illustrates the RUL prediction trajectories for four representative batteries (B0005, B0006, B0007, B0018) from the NASA dataset. The plots display the actual State of Charge (SOC) degradation over discharge cycles, our model's predicted SOC trajectory, and the critical end-of-life (EOL) threshold set at 70%. As shown, our model's predictions closely follow the actual degradation trend, capturing both the gradual capacity fade and the more pronounced regeneration phenomena observed after periods of rest. This ability to track the non-linear and fluctuating nature of battery degradation is a testament to the model's capacity to learn complex temporal patterns.

Adjacency matrices visualization

A key component of our proposed framework is the ability to dynamically learn the correlations between different battery parameters over time. Figures 5 and 6 visualize the learned adjacency matrices for different patches which representing different stages of the battery's life from the NASA and CALCE datasets, respectively. These matrices represent the directed correlation strength between parameters as learned by our Attention-Based Graph Structure Learning module.

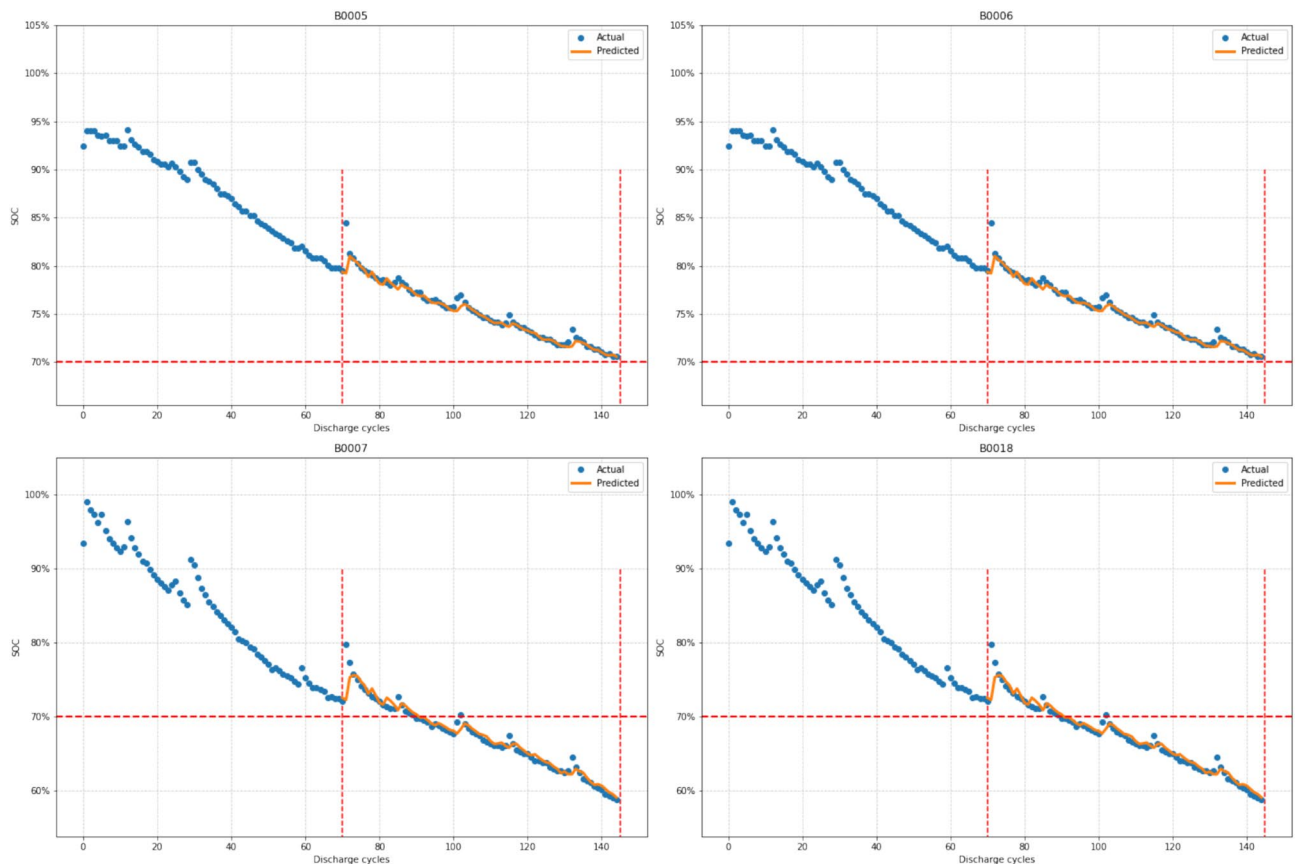
Several key insights can be drawn from these visualizations. Firstly, we observe that the learned correlations are dynamic and evolve as the battery degrades. For instance, in Fig. 5, the learned correlations among Output Current, Loaded Voltage, and Temperature across randomly sampled patches reveal that the model consistently captures key physical relationships. A strong positive correlation between Loaded Voltage and Temperature reflects the expected thermal-electrical coupling, while a moderate correlation between Output Current and Loaded Voltage corresponds to the normal charge-discharge behavior of the cell. In Fig. 6, we can see a consistently strong negative correlation between Internal Resistance and Discharge Energy, as higher resistance leads to greater energy loss. The ability of our model to automatically discover and adapt these meaningful physical relationships from data is a significant advantage for accurate prognostics.

R-squared visualization

To further evaluate the predictive accuracy of our model against state-of-the-art baselines, we present scatter plots of predicted capacity versus measured capacity in Fig. 7. The plots include results from DTNN, MC-CNN-

| Algorithm                     | CALCE dataset |              |              | NASA dataset |              |              | UNIBO dataset |              |              |
|-------------------------------|---------------|--------------|--------------|--------------|--------------|--------------|---------------|--------------|--------------|
|                               | RMSE          | MAE          | MAPE (%)     | RMSE         | MAE          | MAPE (%)     | RMSE          | MAE          | MAPE (%)     |
| MC-CNN-TimesNet <sup>46</sup> | 0.033         | 0.025        | 0.027        | 0.026        | 0.017        | 0.020        | 0.074         | 0.038        | 0.045        |
| DTNN <sup>32</sup>            | 0.039         | 0.028        | 0.030        | 0.033        | 0.019        | 0.022        | 0.082         | 0.041        | 0.049        |
| Conditional GCN <sup>47</sup> | 0.079         | 0.042        | 0.053        | 0.041        | 0.027        | 0.029        | 0.110         | 0.056        | 0.067        |
| T-PATCHGNN <sup>23</sup>      | 0.031         | 0.020        | 0.023        | 0.020        | 0.011        | 0.013        | 0.066         | 0.034        | 0.040        |
| Bat-T-GNN                     | <b>0.024</b>  | <b>0.011</b> | <b>0.012</b> | <b>0.014</b> | <b>0.007</b> | <b>0.008</b> | <b>0.054</b>  | <b>0.027</b> | <b>0.032</b> |

**Table 1.** RUL prediction performance comparison on CALCE, NASA, and UNIBO Datasets. Significant values are in bold.



**Fig. 3.** Case study from the NASA dataset (prediction from 70 cycles).

TimesNet, the original T-PATCHGNN, and our proposed model. The R-squared ( $R^2$ ) value is provided for each model.

The visual results clearly demonstrate the superior performance of our approach. While all models show a positive correlation, the data points for our model are most tightly clustered around the ideal prediction line ( $y=x$ ), indicating lower prediction error. This is quantitatively confirmed by the R-squared values. Our model achieves an impressive  $R^2$  of 0.9725, significantly higher than DTNN (0.8379), MC-CNN-TimesNet (0.8918), and even the original T-PATCHGNN (0.9201). This substantial improvement in  $R^2$  highlights our model's enhanced ability to capture the true degradation behavior.

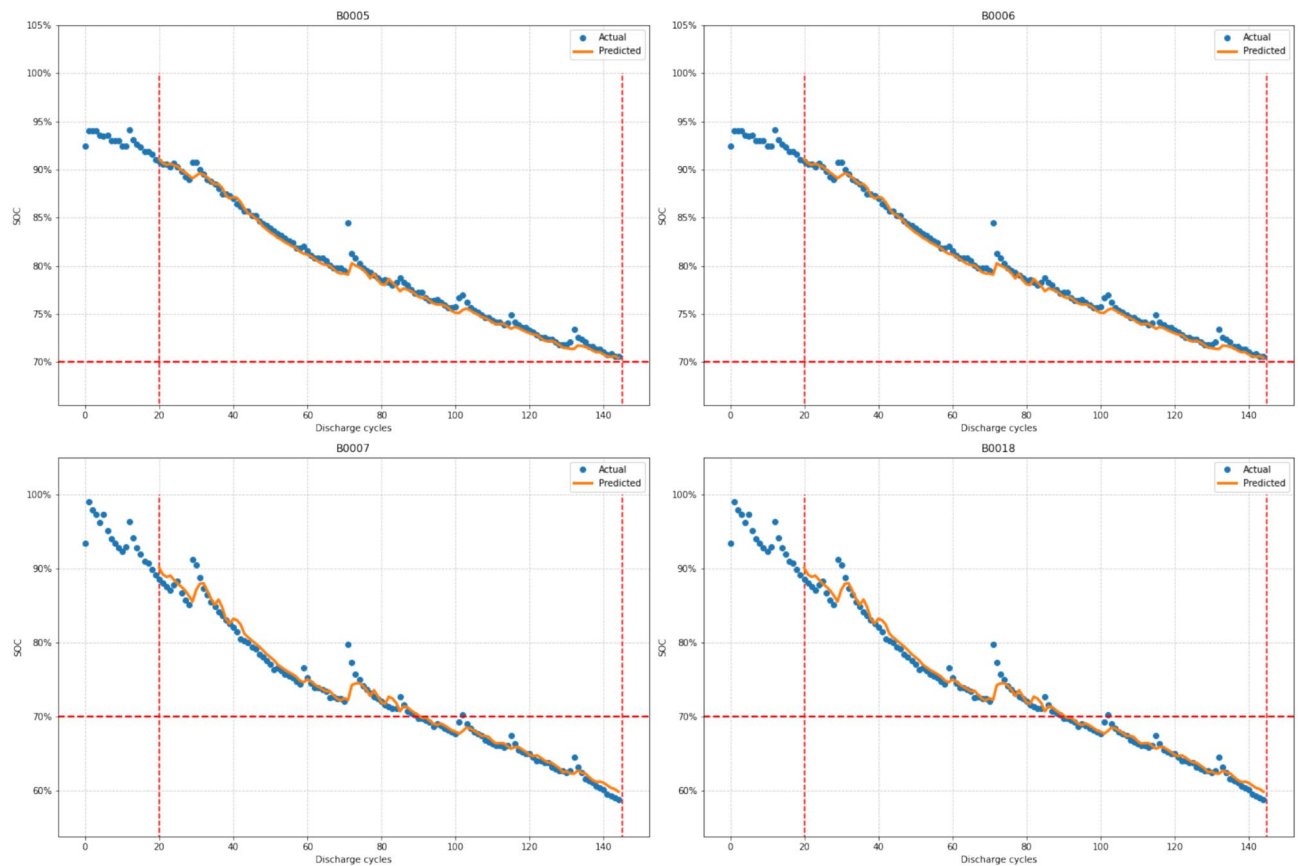
### Hyperparameter sensitivity analysis

To evaluate the robustness of Bat-T-GNN and better understand the influence of our proposed PINN-RUL loss function, we conducted a sensitivity analysis on its key hyperparameters. A model's stability with respect to its key parameters is crucial for its practical applicability, as it indicates that its performance is not the result of a fragile or difficult-to-find configuration. We focused our analysis on the two weighting coefficients,  $\lambda_{\text{phys}}$  and  $\lambda_{\text{cons}}$ , as they directly control the balance between the primary data-driven RUL objective and the physics-informed regularization terms.

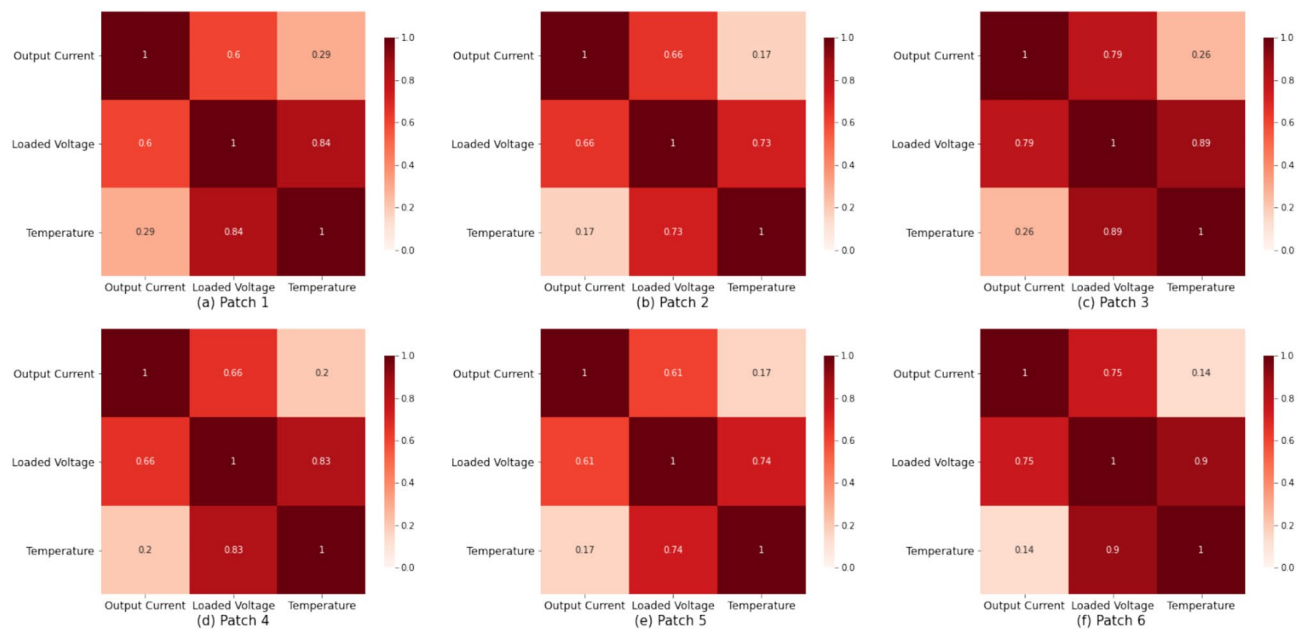
The analysis was performed on the CALCE dataset. We varied one hyperparameter at a time across a predefined logarithmic range while keeping all other parameters fixed at their optimal values, as determined on the validation set. Specifically, we evaluated values of [0.01, 0.1, 1.0, 10.0] for both  $\lambda_{\text{phys}}$  and  $\lambda_{\text{cons}}$ . Performance was measured using the RMSE to assess the impact of these changes.

The results of our sensitivity analysis are presented in Fig. 8. Both plots exhibit a desirable U-shaped curve, which indicates that our model is robust across a reasonable range of hyperparameter values. As observed in the left panel, the model's performance is sensitive to the value of  $\lambda_{\text{phys}}$ . When the weight is too small, the benefit of the physics-informed shape loss is minimal, and the performance is suboptimal. As the weight increases to a moderate value, the RMSE decrease significantly, demonstrating that the physics-based regularization is effectively guiding the model. However, when the weight becomes too large, the performance degrades, suggesting that an overly strong physics constraint can interfere with the primary RUL prediction task.

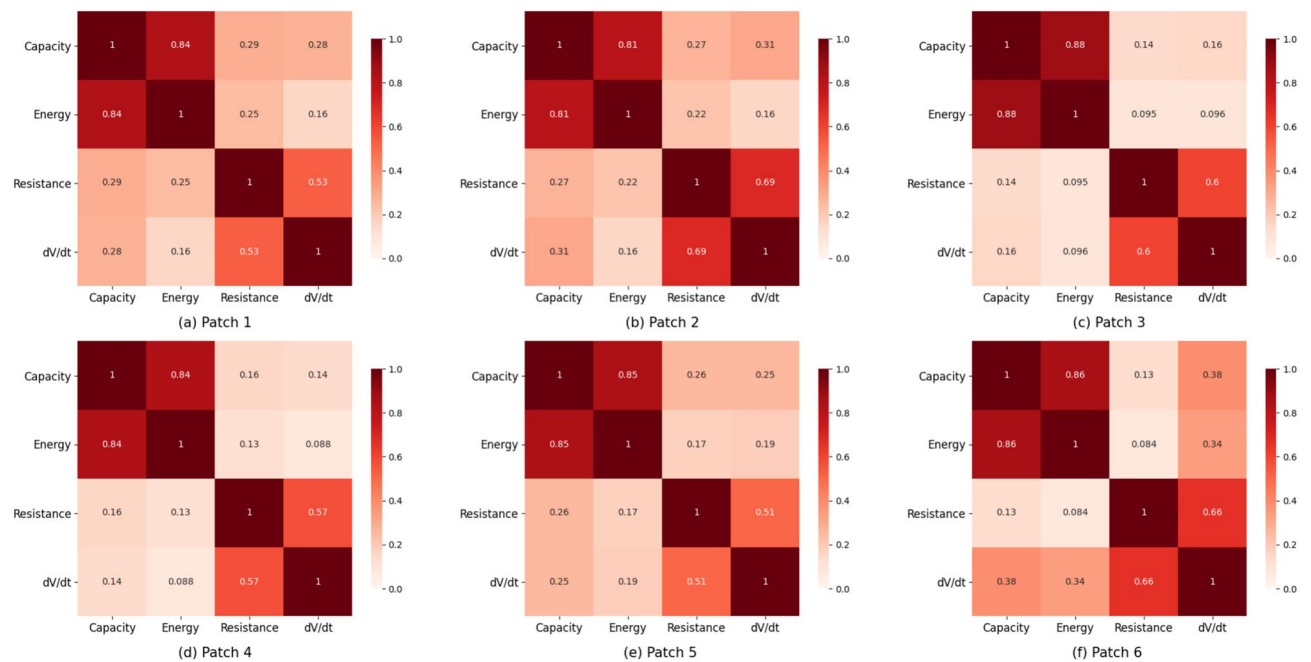
A similar trend is observed for the consistency loss weight,  $\lambda_{\text{cons}}$ , in the right panel of Fig. 8. A very small weight provides little regularization, while a very large weight forces an overly strict agreement between the two prediction heads, potentially harming the main prediction task. The optimal performance is achieved in the range of 0.1 to 1.0, where the consistency loss acts as an effective regularizer without overpowering the learning process. Overall, this analysis demonstrates that while the performance of Bat-T-GNN benefits significantly



**Fig. 4.** Case study from the NASA dataset (prediction from 20 cycles).



**Fig. 5.** Visualization of learned adjacency matrices from the NASA dataset. Parameters includes output current, loaded voltage and temperature.



**Fig. 6.** Visualization of learned adjacency matrices from the CALCE dataset. Parameters includes internal resistance,  $dV/dt$ , discharge energy, and charge capacity.

from our proposed loss components, it is not critically dependent on a highly specific or fine-tuned parameter setting. The model remains stable and performs well across an order of magnitude for both hyperparameters, which supports its robustness and practical applicability.

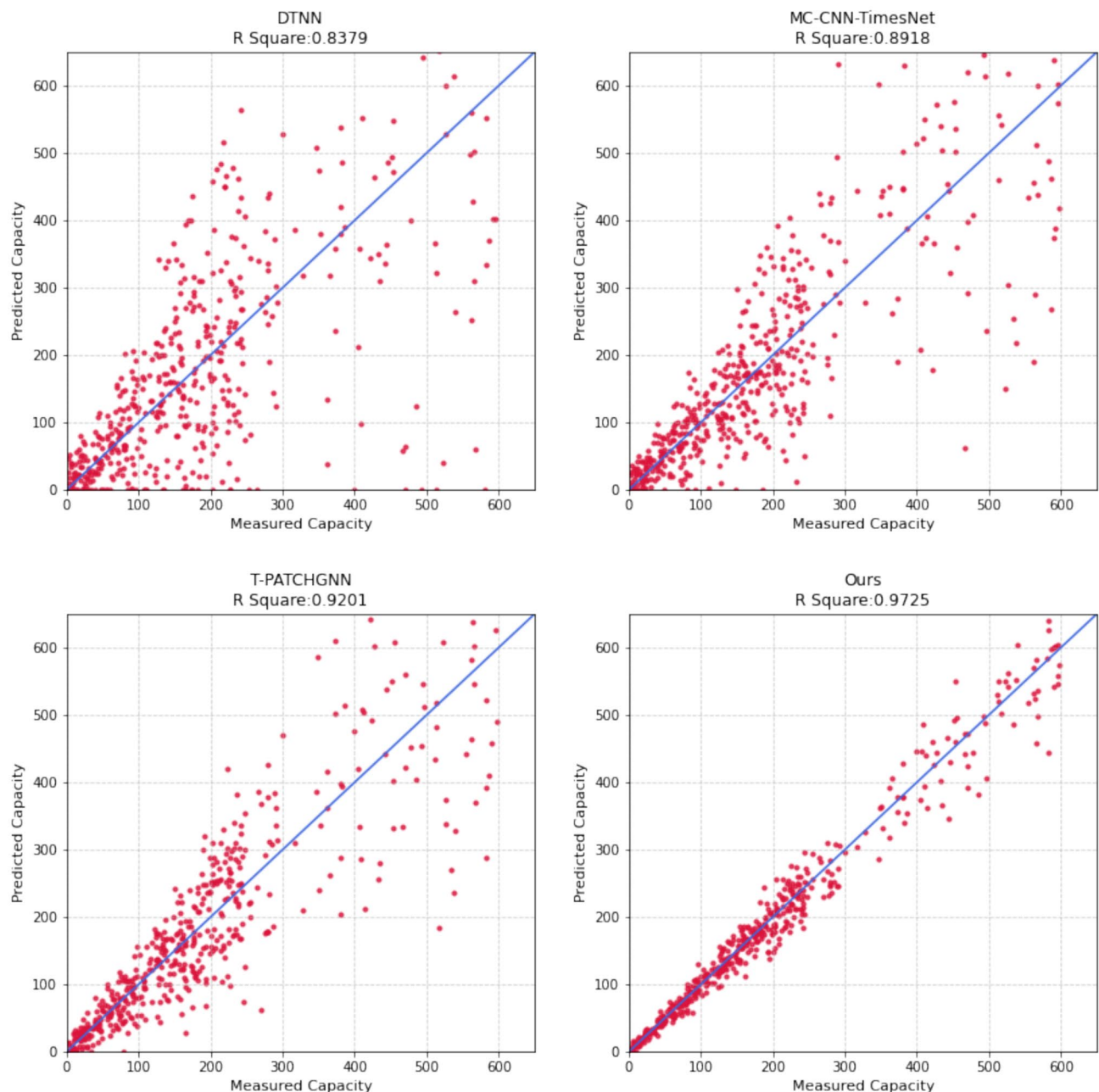
### Ablation analysis

To evaluate and validate the individual contributions of our key innovations—the Cycle-Aware Patching strategy and the Physics-Informed Consistency Loss (PINN-RUL)—we conducted a comprehensive ablation study. The primary goal of this study is to deconstruct our full Bat-T-GNN model to quantify the performance impact of each component. By replacing our proposed modules with more standard alternatives, we can isolate their specific benefits their necessity for achieving the model's state-of-the-art performance.

We created and tested several different versions of our model. Our full Bat-T-GNN, which includes all our new ideas, was used as the main comparison point. The variant A “w/o Cycle-Aware Patching,” used a generic time-window patching method instead of our special cycle-based. This test shows how important it is to organize the data into physical cycles. Variant B “w/o PINN-RUL Loss,” used a standard MSE loss function instead of our new PINN-RUL loss, which tells us how much our physics-based training helps. We also tested Variant C and D that replaced the Transformer with a simpler LSTM network and completely removed the GNN to show that these parts are necessary at inter and intra-relationship construction.

In addition, we designed one more experiment in Table 3 to answer a question: Does our Cycle-Aware Patching strategy improve the performance of the baseline T-PATCHGNN architecture it was built upon? The result clearly shows that applying Cycle-Aware Patching significantly improves the performance of the T-PATCHGNN baseline. The RMSE improved by 15% simply by changing how the input data was segmented. This strongly validates our core hypothesis: providing the model with physically meaningful, semantically complete data segments corresponding to a full operational event reduces the learning burden and prevents the model from being confused by arbitrarily fragmented inputs.

The results from our study are shown in Table 2. The full Bat-T-GNN model clearly performs the best across all metrics, with a total of 48.3 million parameters. We noticed the heavy drop in performance when we removed the Cycle-Aware Patching strategy; the RMSE error increased by more than 58%. This result is significant because it proves our main idea: organizing the data according to the battery's physical cycles is the most critical factor for getting accurate predictions. When the data is split by simple time windows, the model gets a more confusing signal patches, which decrease its ability of feature representation. The version trained with a standard MSE loss also performed worse, which shows the clear benefit of our PINN-RUL loss. This loss function helps guide the model to learn the correct physical degradation path, not just the final RUL number, which makes it more accurate. The other test Variant C and D also showed suboptimal performance, which confirms the importance of the Transformer for long-range dependency modeling and the GNN for capturing dynamic multivariate correlations.

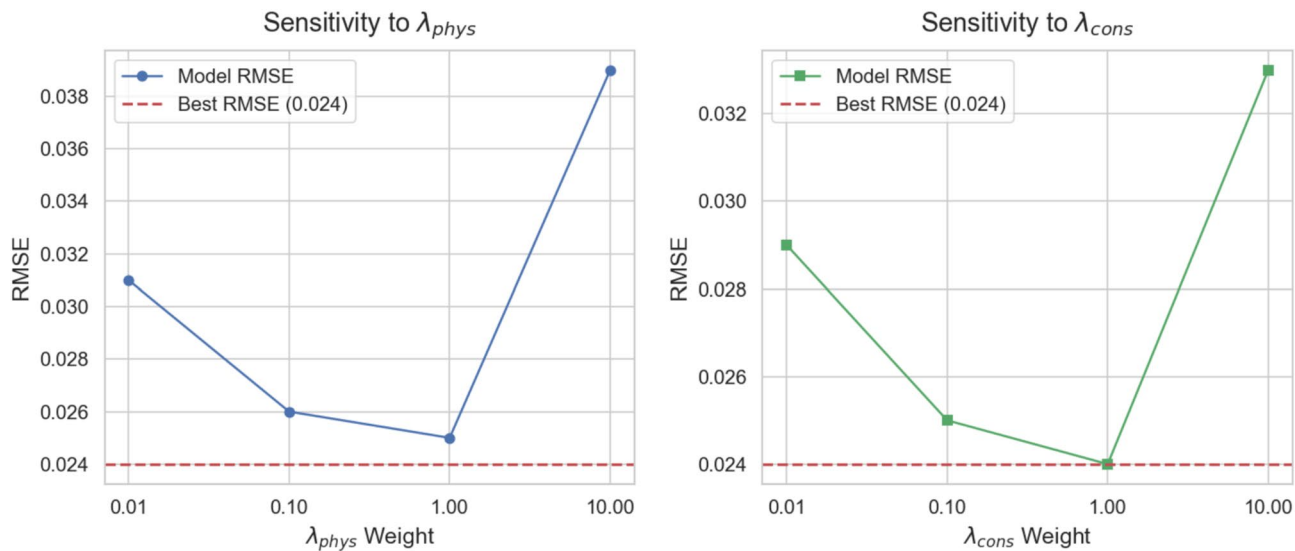


**Fig. 7.** Comparative plots of different models. The higher R-squared value and tighter clustering of points for our model demonstrate its superior predictive accuracy.

### Robustness evaluation

To evaluate real-world deployability on field Battery Management Systems (BMS) data, we stress-test the model under three common degradations on the NASA battery dataset. Three representative perturbations are considered: (i) additive Gaussian noise applied to voltage, current, and temperature channels at SNR = 40 – 10 dB; (ii) random missingness (MCAR) at rates of 5 %, 10 %, and 20 % with simple forward filling; and (iii) slow sensor drift, simulated as a low-frequency bias gradually shifting the measurements. For each condition, we compare the baseline model without the physics-informed loss (w/o PINN-RUL) against the full model including the physics consistency constraint (+ PINN-RUL). Table 4 summarizes the averaged RMSE, MAE, and consistency loss results.

As expected, performance degrades as the signal-to-noise ratio decreases. Nevertheless, the full model consistently maintains lower error across all SNR levels. At 40 dB, RMSE is 0.016 compared with 0.032 for the baseline, and at 10 dB, RMSE remains 0.041 versus 0.065. The MAE shows a similar trend (0.009 vs. 0.019 at 40 dB, and 0.029 vs. 0.041 at 10 dB). The consistency loss also rises with heavier noise but stays bounded (0.35 at 10 dB), showing that the model preserves coherence between data-driven and physics-guided RUL estimates.



**Fig. 8.** Sensitivity analysis of the PINN-RUL loss weighting coefficients  $\lambda_{phys}$  (left) and  $\lambda_{cons}$  (right) on the CALCE dataset. The y-axis represents the RMSE (lower is better). The plots show that performance is robust across a moderate range for both parameters, with optimal values around 0.1 to 1.0.

| Model variant | Description              | RMSE         | MAE          | MAPE         | Param.      | Infer time  | GPU mem     |
|---------------|--------------------------|--------------|--------------|--------------|-------------|-------------|-------------|
| Bat-T-GNN     | Full proposed model      | <b>0.024</b> | <b>0.011</b> | <b>0.012</b> | <b>48.3</b> | <b>25.1</b> | <b>1552</b> |
| Variant A     | w/o Cycle-aware patching | 0.038        | 0.025        | 0.027        | 48.3        | 25.0        | 1552        |
| Variant B     | w/o PINN-RUL loss        | 0.029        | 0.017        | 0.018        | 48.1        | 24.8        | 1539        |
| Variant C     | w/o Transformer          | 0.035        | 0.022        | 0.024        | 45.7        | 13.5        | 1375        |
| Variant D     | w/o GNN                  | 0.033        | 0.020        | 0.021        | 46.5        | 22.1        | 1414        |

**Table 2.** Ablation study results. Performance decrease upon removing or replacing key components highlights their individual contributions. The unit of parameter size is millions (M), the inference time is ms, and the peak gpu memory usage is MB. Significant values are in bold.

| Model            | Patching strategy | RMSE         | MAE          | MAPE         |
|------------------|-------------------|--------------|--------------|--------------|
| T-PATCHGNN       | Generic           | 0.031        | 0.020        | 0.023        |
|                  | Cycle-Aware       | <b>0.027</b> | <b>0.019</b> | <b>0.020</b> |
| Bat-T-GNN (ours) | Generic           | 0.038        | 0.025        | 0.027        |
|                  | Cycle-Aware       | <b>0.024</b> | <b>0.011</b> | <b>0.012</b> |

**Table 3.** Performance comparison isolating the effect of the patching strategy on the CALCE dataset. Significant values are in bold.

When randomly dropping up to 20 % of time-series samples, the full model again demonstrates higher resilience. Its RMSE increases moderately from 0.015 to 0.031 as missingness grows, while the baseline deteriorates from 0.033 to 0.052. The physics constraint effectively compensates for temporal gaps by leveraging physical consistency between successive cycles, reducing the propagation of imputation errors.

Under progressive measurement bias, the advantage of the physics-guided model becomes more pronounced. For high-drift conditions, RMSE decreases from 0.058 to 0.038 and MAE from 0.037 to 0.025 when the PINN-RUL loss is applied. The corresponding reduction in consistency loss (from 0.31 to 0.15–0.20 at moderate levels) indicates that the physics prior counteracts systematic bias accumulation, maintaining alignment between predicted and physically inferred RUL trajectories.

Across all stressors, adding the physics consistency head reduces both error magnitude and error growth rate as conditions worsen, while simultaneously tightening the agreement between predictions and physics. Practically, we recommend enabling the physics head with a moderate consistency weight and pairing it with lightweight data-quality safeguards (robust scaling, drift monitors, and simple imputation). In settings with known high noise or frequent dropouts, the consistency residual itself can serve as an on-line health indicator and domain-shift alarm.

| Condition            | Level  | RMSE         |              | MAE          |              | Consistency loss |
|----------------------|--------|--------------|--------------|--------------|--------------|------------------|
|                      |        | w/o PINN-RUL | + PINN-RUL   | w/o PINN-RUL | + PINN-RUL   | -                |
| Clean (reference)    | –      | 0.021        | <b>0.014</b> | 0.015        | <b>0.007</b> | 0.08             |
| Additive noise (SNR) | 40 dB  | 0.032        | <b>0.016</b> | 0.019        | <b>0.009</b> | 0.11             |
|                      | 30 dB  | 0.038        | <b>0.020</b> | 0.023        | <b>0.012</b> | 0.14             |
|                      | 20 dB  | 0.049        | <b>0.028</b> | 0.030        | <b>0.018</b> | 0.21             |
|                      | 10 dB  | 0.065        | <b>0.041</b> | 0.041        | <b>0.029</b> | 0.35             |
| Random missingness   | 5%     | 0.033        | <b>0.017</b> | 0.020        | <b>0.010</b> | 0.12             |
|                      | 10%    | 0.040        | <b>0.022</b> | 0.025        | <b>0.014</b> | 0.16             |
|                      | 20%    | 0.052        | <b>0.031</b> | 0.033        | <b>0.021</b> | 0.25             |
| Slow sensor drift    | Low    | 0.035        | <b>0.019</b> | 0.022        | <b>0.011</b> | 0.15             |
|                      | Medium | 0.044        | <b>0.026</b> | 0.028        | <b>0.016</b> | 0.20             |
|                      | High   | 0.058        | <b>0.038</b> | 0.037        | <b>0.025</b> | 0.31             |

**Table 4.** Robustness evaluation under controlled data degradation on the NASA dataset. We compare the model without the physics-informed loss (w/o PINN-RUL) against the full model (+ PINN-RUL). Significant values are in bold.

Conclusion

In this paper, we addressed the critical challenge of accurately predicting the Remaining Useful Life (RUL) of Li-ion batteries from irregular, multivariate sensor data. We proposed Bat-T-GNN, a novel framework designed to solve this problem by integrating domain-specific physical knowledge with a state-of-the-art deep learning architecture. This work includes two contributions: first, a Cycle-Aware Patching strategy that aligns data segmentation with the battery’s physical charge-discharge cycles, providing the model with semantically meaningful inputs. Second, we introduced a novel Physics-Informed Consistency Loss (PINN-RUL), which regularizes the training process by ensuring the model’s final prediction is consistent with the physical degradation trajectory it learns. Our extensive experiments on public benchmark datasets demonstrated that Bat-T-GNN sets a new state-of-the-art in RUL prediction, outperforming a wide range of baseline models. Crucially, our ablation studies confirmed that both the Cycle-Aware Patching strategy and the PINN-RUL loss provide significant performance gains, validating their effectiveness as key contributions. The results show that by making a generic SOTA model aware of the problem’s underlying physics, we can achieve a new level of accuracy and robustness.

Limitations and future work

While the proposed Bat-T-GNN framework demonstrates significant advancements in integrating physical knowledge with deep learning for battery RUL prediction, several assumptions and limitations should be acknowledged to guide its real-world adaptation.

First, The PINN-RUL loss currently assumes that degradation follows a smooth and monotonic trend, approximated by an exponential decay model. Although this constraint ensures physical plausibility, real-world degradation often deviates from such ideal forms due to irregular stress events, partial regeneration, or sudden aging phenomena. This may introduce bias when the observed behavior violates the assumed curve. Future work can design adaptive or hybrid physics-informed priors—for instance, by embedding neural ODEs or spline-based degradation functions—so that the physical constraint evolves with data characteristics while maintaining physical interpretability.

Second, The framework assumes consistent scaling across sensor variables, yet real BMS differ in calibration, sampling resolution, and measurement range. Such discrepancies may distort feature magnitudes and impair transferability. Incorporating domain alignment and adaptive normalization strategies, such as feature-wise modulation or adversarial domain adaptation, would enable the model to harmonize signals across platforms and remain robust to hardware-dependent differences.

Thrid, Cycle-Aware Patching relies on current-based thresholds to delineate charge-discharge events. In real applications, especially in electric vehicles or hybrid systems, regenerative braking and partial cycling obscure these boundaries, reducing segmentation reliability. Developing self-supervised or online cycle boundary detection mechanisms—such as sequence clustering or reinforcement-learning-based segmentation—can provide adaptive and noise-tolerant cycle identification, improving patch coherence for industrial data streams.

In addition, The current dynamic graph models only internal battery variables, capturing intrinsic electrochemical relations but overlooking exogenous influences like ambient temperature, load variability, or charging protocol. Extending the graph to include external and environmental nodes would offer a more comprehensive understanding of degradation mechanisms. Multi-modal graph fusion or conditional edge weighting could allow the model to distinguish intrinsic degradation from environmental stressors.

Lastly, extending the model across different chemical systems is an important next step. In principle, the proposed architecture is chemistry-agnostic, since it learns degradation representations from operational time-series while the physics-informed loss enforces consistency in a normalized degradation domain. Nonetheless, the parameters and degradation dynamics of Li-ion chemistries (e.g., NMC, LCO) differ significantly from LFP, NCA, or solid-state systems, and from pack-level multi-cell configurations. Future work will therefore include

cross-chemistry transfer experiments (e.g., LFP → NMC) to evaluate adaptability, and hybrid integration with mechanistic submodels such as SEI-growth, lithium plating, or stress-induced resistance evolution. These enhancements would enable the model to capture chemistry-specific degradation signatures while retaining the data-driven generalization ability demonstrated here.

## Data availability

Experimented on 3 datasets. All data are open access: NASA: “<https://data.nasa.gov/dataset/li-ion-battery-aging-datasets>” UNIBO: “<https://data.mendeley.com/datasets/n6xg5fzsbv/1>” CALCE: “<https://calce.umd.edu/battery-data>”

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## Author contributions

X. C did acquisition, analysis, or interpretation of data. X. C and Y. W did creation of new software used in the work. X. C wrote the main manuscript text. All authors reviewed the manuscript.

## Declarations

## Competing interests

The authors declare no competing interests.

## Additional information

**Correspondence** and requests for materials should be addressed to C.L.

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